A closed-form biomarker-treatment interaction estimator for the simplest hybrid N-of-1 design under compound-symmetric and AR(1) within-subject correlation

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1 Introduction

The estimand of interest in a biomarker-validation N-of-1 program is the biomarker-treatment interaction: the degree to which a participant's response to active drug, relative to its withdrawal, varies with a continuous baseline biomarker. The two preceding manuscripts characterize this estimand by simulation across test procedures and design parameters. This note

isolates its analytic core. For the simplest hybrid design we show that the interaction point estimate has the same closed form regardless of the within-subject correlation structure, while its sampling variance, and therefore its standard error and power, admits distinct closed forms under compound symmetry, under continuous-time AR(1) correlation, and under the other correlation structures in common use, all special cases of a single block-sum identity. The ratio of the compound-symmetric and AR(1) variances quantifies, in a single expression, the efficiency loss that motivates modeling the autocorrelation directly rather than assuming it away.

The design class matters for what the estimand even means, so we fix the distinction at the outset. A conventional cross-over trial randomizes each participant to a treatment sequence (for example AB or BA) and observes one period per treatment; the analysis aggregates the within-participant treatment differences across participants to estimate a single population-average treatment effect, and the participant-by-treatment interaction is treated as a nuisance that is either assumed absent or, in replicate cross-over designs, recovered only as a variance component[1,2]. An aggregated N-of-1 trial instead cycles each participant through the on-treatment and off-treatment states repeatedly, so that the individual treatment effect is identified within each participant before any aggregation[3,4]. Pooling those individual effects in a mixed model yields not only the population average but the distribution of individual effects, and, when a biomarker is measured, the biomarker-treatment interaction that is the predictive-medicine quantity of interest[5]. The hybrid design of[6] is the specialisation we study: an open-label run-in places every participant on drug, after which blinded discontinuation moves them off drug, generating the within-participant on/off contrast that carries the interaction while avoiding the placebo-run-in selection bias of a response-gated design. In short, a cross-over trial spends its within-participant information on the average effect; an aggregated N-of-1 trial spends it on the individual effect and its modification by the biomarker.

We work with the minimal hybrid that still supports the interaction contrast, derive the estimator and its variance under both correlation structures, give the efficiency ratio and a worked example, and finally restore the open-label phase to show that the same closed-form machinery yields separate pharmacological and expectation interaction estimators.

2 The simplest hybrid design

Consider N participants indexed $i = 1, \dots, N$, each with a continuous, centered baseline biomarker b_i (so $\sum_i b_i = 0$). Each participant is observed at $T = p + q$ equally spaced occasions: the first p occasions on drug (the open-label run-in, $D_t = 1$, $t = 1, \dots, p$) and the remaining q off drug (after blinded discontinuation, $D_t = 0$, $t = p + 1, \dots, p + q$). This is the minimal hybrid that retains a within-participant on/off contrast, and hence the only structure needed to identify the biomarker-treatment interaction; the two-path Hendrickson hybrid, in which participants discontinue at different occasions, is a refinement that we do not require here. Figure 1 shows the layout at the reference cycling depth $p = q = 4$.

The data-generating model is the additive fixed-effects form

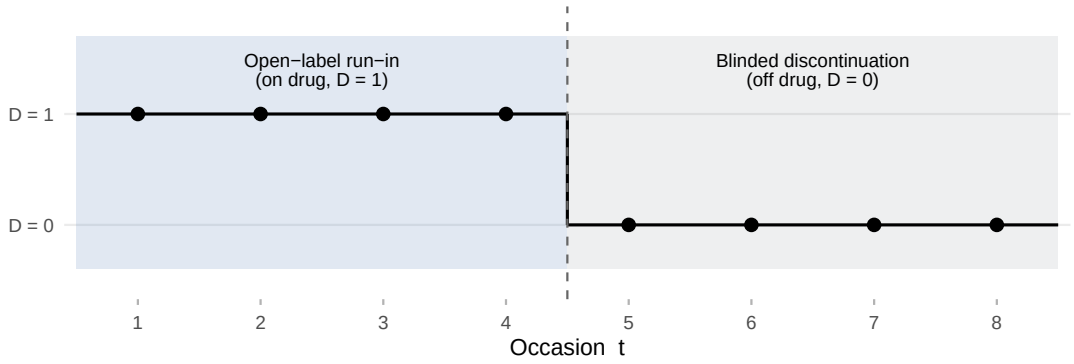


Figure 1: The simplest hybrid design with $p = 4$ on-drug and $q = 4$ off-drug occasions. Every participant begins on active drug (open-label run-in, drug indicator $D_t = 1$) and is then taken off drug under blinding (discontinuation, $D_t = 0$) at the dashed transition. Filled points mark the measurement occasions; the within-participant on/off contrast of these occasions, regressed on the biomarker across participants, identifies the interaction θ.

$$y_{it} = \mu + \beta\, b_i + \gamma\, D_t + \theta\,(b_i D_t) + e_{it}, \tag{1}$$

where θ is the biomarker-treatment interaction (the estimand), and the within-participant errors $e_i = (e_{i1}, \ldots, e_{iT})^\top$ are independent across participants with common marginal variance $\mathrm{Var}(e_{it}) = \sigma^2$ and a within-participant correlation matrix R to be specified. In the project's data-generating process θ equals the moderation strength times the response scale, $\theta = \beta_{bm}\,\sigma_{BR}$ (for the prazosin-PTSD reference, $0.45 \times 5 = 2.25$).

3 The interaction estimator

Form each participant's within-participant treatment contrast by averaging within phase and differencing across phase,

$$\Delta_i = \bar{y}_i^{\mathrm{on}} - \bar{y}_i^{\mathrm{off}} = \frac{1}{p}\sum_{t=1}^{p} y_{it} - \frac{1}{q}\sum_{t=p+1}^{p+q} y_{it}. \tag{2}$$

The intercept μ, the biomarker main effect βb_i, and any participant-level offset are constant across t within a participant and so cancel in (2), leaving

$$\Delta_i = \gamma + \theta\, b_i + \varepsilon_i, \qquad \varepsilon_i = \frac{1}{p}\sum_{t=1}^{p} e_{it} - \frac{1}{q}\sum_{t=p+1}^{p+q} e_{it}. \tag{3}$$

Equation (3) is an ordinary simple regression of the contrast Δ_i on the biomarker b_i, so the interaction estimator is the least-squares slope

$$\hat{\theta} = \frac{\sum_{i=1}^{N} (b_i - \bar{b})\, \Delta_i}{\sum_{i=1}^{N} (b_i - \bar{b})^2}, \qquad \mathbb{E}[\hat{\theta}] = \theta. \tag{4}$$

Two features of (4) are worth stating explicitly. First, the point estimate is unbiased and has the same closed form for any within-participant correlation structure: the structure enters only through ε_i, which is mean zero and does not touch the numerator's expectation. Second, because every participant shares the same design, the contrasts ε_i are homoscedastic across i and independent between participants, so ordinary least squares is fully efficient (generalized least squares coincides with it) and

$$\mathrm{Var}(\hat{\theta}) = \frac{V_\varepsilon}{\sum_i (b_i - \bar{b})^2} = \frac{V_\varepsilon}{(N-1)\, s_b^2}, \qquad V_\varepsilon \equiv \mathrm{Var}(\varepsilon_i), \tag{5}$$

with s_b^2 the sample variance of the biomarker. The entire effect of the correlation structure is concentrated in the single scalar V_ε, which we now evaluate under each structure.

4 Variance under compound symmetry

Under compound symmetry every distinct pair of within-participant occasions shares the same correlation, $\mathrm{Cor}(e_{it}, e_{is}) = \rho$ for $t \neq s$. The two phase means have variances

$$\mathrm{Var}(\bar{y}_i^{\mathrm{on}}) = \sigma^2\Big(\rho + \tfrac{1-\rho}{p}\Big), \qquad \mathrm{Var}(\bar{y}_i^{\mathrm{off}}) = \sigma^2\Big(\rho + \tfrac{1-\rho}{q}\Big), \tag{6}$$

and, since every on/off pair shares the same covariance $\rho\sigma^2$, their cross-covariance is $\mathrm{Cov}(\bar{y}_i^{\mathrm{on}}, \bar{y}_i^{\mathrm{off}}) = \rho\sigma^2$. Substituting into $V_\varepsilon = \mathrm{Var}(\bar{y}_i^{\mathrm{on}}) + \mathrm{Var}(\bar{y}_i^{\mathrm{off}}) - 2\,\mathrm{Cov}(\cdot)$ cancels the shared ρ terms and gives the compact result

$$\boxed{V_\varepsilon^{\mathrm{CS}} = \sigma^2(1-\rho)\Big(\tfrac{1}{p} + \tfrac{1}{q}\Big).} \tag{7}$$

The interaction variance therefore depends on the correlation only through the common factor $(1-\rho)$: the shared between-occasion component cancels in the within-participant contrast, and what remains is the independent innovation $\sigma^2(1-\rho)$ averaged over the p and q occasions. The arrangement of occasions is irrelevant under compound symmetry; only the counts p and q enter.

5 Variance under AR(1)

Under continuous-time AR(1) correlation on equally spaced occasions, $\mathrm{Cor}(e_{it}, e_{is}) = \varphi^{|t-s|}$ with $0 \leq \varphi < 1$, the off-diagonal covariances decay with separation, so the arrangement of

occasions now matters. The sum of all entries of an $m \times m$ AR(1) correlation matrix has the closed form

$$S_m \equiv \mathbf{1}^\top R_m \mathbf{1} = \frac{m(1+\varphi)}{1-\varphi} - \frac{2\varphi\,(1-\varphi^m)}{(1-\varphi)^2}, \tag{8}$$

so that $\mathrm{Var}(\bar{y}_i^{\mathrm{on}}) = \sigma^2 S_p/p^2$ and $\mathrm{Var}(\bar{y}_i^{\mathrm{off}}) = \sigma^2 S_q/q^2$. Because the on-drug block ($t = 1, \dots, p$) and the off-drug block ($t = p+1, \dots, p+q$) are contiguous, the cross-block covariance sum factorises,

$$\sum_{t=1}^{p} \sum_{s=p+1}^{p+q} \varphi^{\,s-t} = \frac{(1-\varphi^p)}{1-\varphi} \cdot \frac{\varphi\,(1-\varphi^q)}{1-\varphi} \;\equiv\; C_{pq}, \tag{9}$$

giving $\mathrm{Cov}(\bar{y}_i^{\mathrm{on}}, \bar{y}_i^{\mathrm{off}}) = \sigma^2 C_{pq}/(pq)$. Assembling the three terms,

$$\boxed{V_\varepsilon^{\mathrm{AR}(1)} = \sigma^2 \left(\frac{S_p}{p^2} + \frac{S_q}{q^2} - \frac{2\,C_{pq}}{pq} \right),} \tag{10}$$

with S_p, S_q, C_{pq} from (8)–(9). The structural contrast with (7) is the cross-block covariance: under compound symmetry the on and off phase means remain strongly correlated however many occasions separate them, so the contrast cancels most of the variance; under AR(1) the contiguous on and off blocks are temporally separated, their covariance C_{pq} decays geometrically, and the contrast benefits from little cancellation. The autocorrelation that helps the phase means (tighter within-block averaging) hurts the contrast (weaker between-phase cancellation), and the latter dominates.

A useful sanity check: with a single occasion per phase, $p = q = 1$, both formulas reduce to $V_\varepsilon = 2\sigma^2(1-\rho)$ and $2\sigma^2(1-\varphi)$ respectively, which coincide when $\rho = \varphi$. With one measurement per phase there is no within-phase averaging, so only the lag-one correlation is identifiable and the two structures are indistinguishable; the structures diverge precisely when p or q exceeds one.

6 A unifying formula and other correlation structures

Equations (7) and (10) are two instances of a single identity. For an arbitrary within-participant correlation matrix R, partition the occasions into the on-drug block $\mathcal{O} = \{1, \dots, p\}$ and the off-drug block $\mathcal{F} = \{p+1, \dots, p+q\}$, and define the three block correlation sums $S_{\mathrm{on}} = \sum_{t,s \in \mathcal{O}} R_{ts}$, $S_{\mathrm{off}} = \sum_{t,s \in \mathcal{F}} R_{ts}$, and $S_\times = \sum_{t \in \mathcal{O},\, s \in \mathcal{F}} R_{ts}$. Because the contrast (2) carries weight $1/p$ on each on-drug occasion and $-1/q$ on each off-drug occasion, $V_\varepsilon = \sigma^2\, a^\top R\, a$ for that weight vector a, which is exactly

$$V_\varepsilon = \sigma^2 \Big(\frac{S_{\mathrm{on}}}{p^2} + \frac{S_{\mathrm{off}}}{q^2} - \frac{2\,S_\times}{pq} \Big). \tag{11}$$

Compound symmetry is the case $S_{\text{on}} = p[1 + (p-1)\rho]$, $S_{\times} = pq\,\rho$; AR(1) is $S_{\text{on}} = S_p$ of (8) and $S_{\times} = C_{pq}$ of (9). The same template delivers the remaining structures in common use.

Toeplitz and banded. If the correlation depends only on the lag, $R_{ts} = \rho_{|t-s|}$ with $\rho_0 = 1$ and free $\rho_1, \rho_2, \ldots$ (AR(1) is the one-parameter case $\rho_k = \varphi^k$; an m-dependent or banded model sets $\rho_k = 0$ for $k > m$; a general stationary, or Toeplitz, model leaves all of them free), the block sums are lag-weighted counts,

$$S_{\text{on}} = p + 2\sum_{k=1}^{p-1}(p-k)\,\rho_k, \qquad S_{\text{off}} = q + 2\sum_{k=1}^{q-1}(q-k)\,\rho_k,$$

$$S_{\times} = \sum_{\ell=1}^{p+q-1} \min(\ell,\, p,\, q,\, p+q-\ell)\;\rho_\ell, \tag{12}$$

where the cross-block weight $\min(\ell, p, q, p+q-\ell)$ counts the on/off pairs separated by lag ℓ. Substituting (12) into (11) gives the interaction-contrast variance for any Toeplitz or banded structure; $\rho_k = \varphi^k$ recovers (10) and $\rho_k = \rho$ recovers (7).

Unstructured. With every correlation free, (11) is itself the answer, $V_\varepsilon = \sigma^2\, a^\top \hat{R}\, a$ with $\hat{R}$ the estimated correlation, but the structure costs $T(T+1)/2$ parameters and needs $N \gg T$ to estimate, which an N-of-1 trial rarely affords; it is the gold standard for MMRM in large parallel-group trials, not here.

Independence and random slopes. Independence ($\rho_k = 0$) gives the naive $V_\varepsilon = \sigma^2(1/p + 1/q)$, the largest contrast variance in the family. A random intercept alone reproduces compound symmetry and cancels in the contrast; adding a random time slope of variance τ^2 contributes a term $\tau^2(\bar{t}_{\text{on}} - \bar{t}_{\text{off}})^2$ to V_ε that does not cancel and grows with the temporal separation of the two phases, a penalty specific to the discontinuation design.

Equations (11)–(12) agree with the full quadratic form $a^\top R\, a$ to within 2e-16 for a representative non-AR(1) Toeplitz correlation, confirming the block-sum bookkeeping.

7 Efficiency and power

Because the point estimate is common to both structures, the entire inferential difference is the variance ratio $V_\varepsilon^{\text{AR(1)}}/V_\varepsilon^{\text{CS}}$ evaluated at a matched marginal variance σ^2 and a matched lag-one correlation $\rho = \varphi$. From (5) the Wald statistic is $\hat{\theta}/\text{SE}(\hat{\theta})$ with $\text{SE}(\hat{\theta}) = \{V_\varepsilon/[(N-1)s_b^2]\}^{1/2}$, so under the alternative the non-centrality is

$$\zeta = \theta\sqrt{\frac{(N-1)\,s_b^2}{V_\varepsilon}}, \qquad \frac{\zeta_{\text{AR(1)}}}{\zeta_{\text{CS}}} = \sqrt{\frac{V_\varepsilon^{\text{CS}}}{V_\varepsilon^{\text{AR(1)}}}}. \tag{13}$$

Power is monotone increasing in ζ, and the relative sample size required to match power is exactly the variance ratio. The following evaluates the two closed forms (and confirms them

Table 1: Interaction-contrast variance under compound symmetry and AR(1) at matched lag-one correlation $\rho = \varphi = 0.5$, balanced phases $p = q$. The ratio is the factor by which the AR(1) design inflates $\mathrm{Var}(\hat{\theta})$, equal to the relative number of participants needed to match power.

	p	q	$V_\varepsilon^{CS}/\sigma^2$	$V_\varepsilon^{AR(1)}/\sigma^2$	ratio (= relative N)
1	1	1	1.000	1.000	1.000
5	4	4	0.250	0.812	3.246
9	8	8	0.125	0.563	4.508

against the full covariance computation).

```
S  <- function(m, phi) m * (1 + phi) / (1 - phi) -
        2 * phi * (1 - phi^m) / (1 - phi)^2
Cc <- function(p, q, phi) phi * (1 - phi^p) * (1 - phi^q) / (1 - phi)^2
V_cs  <- function(p, q, rho) (1 - rho) * (1 / p + 1 / q)
V_ar1 <- function(p, q, phi) S(p, phi) / p^2 + S(q, phi) / q^2 -
           2 * Cc(p, q, phi) / (p * q)

grid <- expand.grid(p = c(1L, 4L, 8L), q = c(1L, 4L, 8L),
                    rho = 0.5)
grid <- grid[grid$p == grid$q, ]
grid$V_CS    <- mapply(V_cs,  grid$p, grid$q, grid$rho)
grid$V_AR1   <- mapply(V_ar1, grid$p, grid$q, grid$rho)
grid$ratio   <- grid$V_AR1 / grid$V_CS
grid$N_infl  <- grid$ratio
knitr::kable(
  grid[, c('p', 'q', 'V_CS', 'V_AR1', 'ratio')],
  col.names = c('$p$', '$q$', '$V^{CS}_\\varepsilon/\\sigma^2$',
                '$V^{AR(1)}_\\varepsilon/\\sigma^2$',
                'ratio (= relative $N$)'),
  digits = 3, escape = FALSE,
  caption = 'Interaction-contrast variance under compound symmetry
    and AR(1) at matched lag-one correlation $\\rho=\\varphi=0.5$,
    balanced phases $p=q$. The ratio is the factor by which the AR(1)
    design inflates $\\operatorname{Var}(\\hat\\theta)$, equal to the
    relative number of participants needed to match power.'
)
```

At the reference cycling depth of four on-drug and four off-drug occasions, with $\rho = \varphi = 0.5$, the AR(1) interaction-contrast variance is 3.25 times the compound-symmetric value (equation (7) gives $0.25\,\sigma^2$, equation (10) gives $0.81\,\sigma^2$). Equivalently, the Wald statistic shrinks by a factor 0.56, and roughly 3.2 times as many participants are needed to recover the power available under compound symmetry. The inflation grows with cycling depth: at $p = q = 8$

the ratio exceeds 4.5, because deeper within-phase averaging buys the compound-symmetric contrast more cancellation than it buys the AR(1) contrast.

8 Including expectation: the three-phase hybrid

The minimal design collapses a feature the hybrid is built around: a participant knows they are on drug during the open-label run-in but is blinded during discontinuation, so belief and exposure coincide in the run-in and decouple thereafter. The same machinery accommodates this with one extra regressor and returns a separate closed-form estimate for each channel.

Partition each participant's occasions into three balanced classes: open-label on-drug (p_1 occasions, exposure $D = 1$, belief $E = 1$), blinded on-drug (p_2, $D = 1$, $E = e_0$), and blinded off-drug (q, $D = 0$, $E = e_0$), where e_0 is the blinded belief level ($e_0 = 1/2$ under 1:1 discontinuation). The analysis model adds an expectation main effect and an expectation-by-biomarker interaction,

$$y_{it} = \mu + \beta\, b_i + \gamma_D D_t + \gamma_E E_t + \theta_D\,(b_i D_t) + \theta_E\,(b_i E_t) + e_{it}, \tag{14}$$

in which θ_D is the biomarker-by-drug (pharmacological) interaction, the estimand, and θ_E is the biomarker-by-expectation (placebo) interaction that an open-label-only analysis would confound with it.

Let m_{1i}, m_{2i}, m_{3i} denote the participant's mean over the three classes. Each is linear in b_i with class-specific slope $s_1 = \beta + \theta_D + \theta_E$, $s_2 = \beta + \theta_D + \theta_E e_0$, and $s_3 = \beta + \theta_E e_0$; the system is non-singular for $e_0 \neq 1$ and gives $\theta_D = s_2 - s_3$ and $\theta_E = (s_1 - s_2)/(1 - e_0)$. Because a difference of slopes is the slope of the difference, each interaction is the least-squares slope of a single within-participant contrast,

$$\boxed{\hat{\theta}_D = \frac{\sum_i (b_i - \bar{b})\,(m_{2i} - m_{3i})}{\sum_i (b_i - \bar{b})^2}} \qquad \text{(blinded-on minus blinded-off; belief held at } e_0\text{)}, \tag{15}$$

$$\hat{\theta}_E = \frac{1}{1 - e_0}\,\frac{\sum_i (b_i - \bar{b})\,(m_{1i} - m_{2i})}{\sum_i (b_i - \bar{b})^2} \qquad \text{(open-label-on minus blinded-on; drug held at 1).} \tag{16}$$

Both are unbiased. The pharmacological interaction θ_D is identified entirely within the blinded window, where belief is fixed at e_0 and so cancels in the contrast; the expectation interaction θ_E is identified by the run-in versus blinded-on contrast, where exposure is fixed at one. The empty fourth cell – off drug while certain of being on – is no obstacle, because the two interactions are read from different contrasts rather than from a crossed factorial.

The variance of the estimand follows the block-sum identity (11) applied to the two blinded blocks,

$$\mathrm{Var}(\hat{\theta}_D) = \frac{V_D}{(N-1)\, s_b^2}, \qquad V_D = \sigma^2 \left(\frac{S_{p_2}}{p_2^2} + \frac{S_q}{q^2} - \frac{2\, C_{p_2 q}}{p_2 q} \right), \tag{17}$$

with the same S_m and C_{pq} as before. Under compound symmetry $S_{p_2} = p_2[1 + (p_2 - 1)\rho]$ and $C_{p_2 q} = p_2 q\, \rho$, so $V_D = \sigma^2(1 - \rho)(1/p_2 + 1/q)$; under AR(1) they are (8)–(9). The drug interaction therefore inherits exactly the compound-symmetry-versus-AR(1) efficiency contrast derived above, now governed by the blinded-window depths p_2 and q rather than the full on/off counts. The minimal design of the preceding sections is the special case in which the open-label block is dropped from the contrast, the belief term is constant, and θ_E vanishes, leaving θ_D as the sole interaction. The two estimators are correlated through the shared mean m_{2i}, with covariance obtained from the same block sums; a joint generalized-least-squares fit recovers the small remaining efficiency, exactly as the full mixed model does relative to the equal-weight contrasts.

9 Remarks

The result makes precise the qualitative claim of the companion manuscripts. A classical analysis that presumes compound symmetry, whether strict repeated-measures ANOVA or a random-intercept model without a residual-correlation term, prices the interaction at $V_\varepsilon^{\mathrm{CS}}$ and so reports a standard error that is too small when the truth is AR(1). The shortfall equals the square root of the variance ratio above only when the compound-symmetry correlation that the analysis uses matches the lag-one AR(1) correlation; a fitted compound-symmetry model estimates an average correlation over all pairs of occasions, which is smaller, so its understatement is smaller than that ratio implies (argued; not computed for this design). (Corrected 2026-10-08: this sentence previously stated the square-root ratio as the understatement without that condition. See `analysis/report/compendium-mathematics.tex`, Section "Design efficiency and time adjustment", subsection "The simplest hybrid design: an exact estimator".) A model that estimates the AR(1) parameter directly (for example `nlme::lme` with `corCAR1`) targets $V_\varepsilon^{\mathrm{AR}(1)}$ and is correctly calibrated. Two extensions follow the same algebra and are left as remarks. First, exponential carryover replaces the binary off-drug indicator D_t by a continuous decay $D_t = \exp(-\lambda\, \mathrm{tsd})$; the contrast weights in (2) change but the variance bookkeeping of (7) and (10) is unchanged in form. Second, dichotomizing the biomarker at its median replaces s_b^2 in (5) by the smaller between-stratum variance, which is the separate, structure-independent power penalty that the test-procedure manuscript isolates.

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